

RAHUL BAMANIYA

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EDUCATION

Shri G.S. Institute of Technology & Science (SGSITS)

Bachelor of Technology in Computer Science and Engineering | CGPA: 7.6

Nov 2022 – Present

Indore, India

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL
Databases: MySQL, MongoDB, Redis
AI / GenAI: LangChain, LangGraph, Prompt Engineering, RAG
Backend: FastAPI, Node.js, REST APIs, Microservices, Swagger/OpenAPI
Tools & Platforms: Git, Docker, GitHub Actions, Linux, Postman
Software Engineering: DSA, OOP, Agile, SDLC, System Design, Testing, Code Review

WORK EXPERIENCE

Generative AI Software Engineer Intern

Zangoh (Newzera Tech Labs Pvt. Ltd.)

Jan 2026 – Present

Indore, India

- Engineered an **agentic media-generation workflow** using **LangGraph** multi-agent orchestration with human-in-the-loop control, automating end-to-end AI content production pipelines.
- Redesigned and migrated existing document preprocessing pipeline to **Kornia** (GPU-accelerated), improving backend REST API performance and reducing processing latency by **98%** (3s → 140ms).
- Maintained **high code quality** through peer code reviews, adherence to clean code principles, and implementation of comprehensive unit and integration test cases using **pytest** and **GitHub Actions** CI/CD.

AI/ML Research Intern

NIT Bhopal, Dept. of Computer Science & Engineering

Jun 2025 – Jul 2025

Bhopal, India

- Fine-tuned a Vision-Language Model on satellite imagery via **QLoRA/LoRA**; integrated **RAG** pipeline with **Mistral LLM** — improved caption quality by **15%**; published at **ICADCML 2026**.
- Built **CLIP**-based vector retrieval and **RAG** pipelines enabling multimodal VQA and semantic reasoning over satellite imagery.

PROJECTS

RouteOptimizer: Last-Mile Logistics Platform | Tech Lead · 5-Member Team

[\(GitHub\)](#)

- Built a **CVRPTW optimization engine** as an independent Python microservice, reducing routing latency by **40%** through load testing and debugging.
- Integrated **OSRM API** for real road-network with QR-based driver route access, achieving **97.1%** accuracy on test data.
- Led 5-member team using **Agile/Scrum** — owned sprint planning, module assignments, and code reviews delivering the platform end-to-end.

Tech Stack: Python, Node.js, OR-Tools, OSRM, Docker, System Design

ContextFlow — Self-Correcting RAG Engine

[\(GitHub\)](#)

- Architected a production-grade **self-correcting RAG engine** with adaptive retrieval, query rewriting, and automated validation workflows, improving response grounding and retrieval quality for document intelligence.
- Designed resilient inference and deployment workflows with failover, monitoring, and deployment automation to improve reliability, availability, and production readiness.

Tech Stack: Python, FastAPI, LangGraph, Qdrant, MongoDB, Redis, Docker, GitHub Actions

PUBLICATION

- *Satellite Image Analysis using RAG & Vision-Language Models (VLMs)* — Accepted at International Conference on Advances in Distributed Computing and Machine Learning (**ICADCML 2026**) (*Under Publication*)

CERTIFICATIONS & ACHIEVEMENTS

- **NPTEL Discipline Star** — IIT Madras (Jan–Apr 2026) — [Certificate](#)
- **AI Foundations Associate** — Oracle Cloud Infrastructure 2025 — [Certificate](#)
- **GSSoC 2024:** Ranked in **Top 1.4%** (240/17,000+) in open source contributions — [GitHub](#)
- **Hacktoberfest 2024:** Contributed to open source with **20+ pull requests** accepted on GitHub
- **400+ DSA Problems Solved** on LeetCode, GeeksforGeeks, and Codeforces